

Danyal Özbek

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EDUCATION

Stockholm University

Bachelor of Science in Computer and Systems Sciences

Stockholm

Aug. 2023 – June 2026

Högskolan i Gävle

Computer Engineering

Gävle

Aug. 2022 – June 2023

BACHELOR'S THESIS

Anomaly Detection on IT Telemetry | Python, PyOD, scikit-learn, Pandas, PyTorch

2026

- Trained, validated and compared three unsupervised anomaly detection models (OCSVM, Isolation Forest, LSTM-Autoencoder) to identify irregularities in multivariate time-series data from IT systems.
- Preprocessed and extracted features from IT telemetry logs, ensuring clean inputs for training.
- Evaluated models with AUC-PR/ROC, precision, recall and F1, analysing the trade-off between missed incidents and false alarms to set a baseline for automated incident detection.

PROJECTS

Essay | Code Analysis & Verification | Kotlin, Compiler PSI/Analysis API, Kotest, Ktor

- Built a code analysis tool that generates property-based tests to verify the correctness of Kotlin projects.
- Parsed functions with the Kotlin Compiler PSI/Analysis API and generated invariants (properties that must always hold) via LLMs (Ktor and Ollama).
- Tested with Kotest; set up GitHub Actions CI/CD as quality gates that catch defects before merge.

Trade Surveillance | Java, Spring Boot, REST API, JUnit

- Built a prototype surveillance application that detects statistical outliers and anomalous trading activity in stock exchange data.
- Developed a RESTful API with Spring Boot to expose detection results to client applications.
- Verified the detection logic with unit and integration tests (JUnit, Spring Boot Test).

Game Engine API | C++, SDL3, Object-Oriented Design

- Designed a reusable C++ API for 2D games, separating general engine functionality from game code.
- Integrated SDL3 behind the engine's own interfaces for rendering, windowing and real-time input.
- Built modular base classes with clear responsibilities, so new game objects are added with minimal boilerplate.

EXPERIENCE

Private Tutor

2024 – 2026

Freelance

Stockholm, Sweden

- Coached university students in programming and CS courses.

Retail Associate & Event Staff

2023 – 2024

Retail Knowledge AB (Blomsterlandet); Five Moments AB

Stockholm, Sweden

- Customer service, sales and cashier work.

SKILLS

Programming: Java, Python, Kotlin, C++, SQL

Frameworks & Tools: Spring Boot, JUnit, Kotest, Ktor, SDL3, PyTorch, scikit-learn, Pandas, PyOD, Git, GitHub Actions, Docker

Concepts: Software Testing, Anomaly Detection, Machine Learning, OOP, Algorithms, Databases, REST